

TAFSIR CHOWDHURY

Full Stack AI Engineer

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Summary

Backend-first Full Stack AI Engineer with system design depth, AI pipeline ownership, and DevOps working knowledge. 3+ years building production-grade web applications with TypeScript, Node.js, Next.js, and cloud infrastructure (AWS EC2/ECS, Railway). Proven track record architecting multi-LLM pipelines (OpenAI, Claude, Gemini), distributed worker systems, and scalable REST APIs for global clients. BSc in Computer Science with a 3.67 GPA and multiple IBM and Meta certifications.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
Frontend: React.js, Next.js, Tanstack Query, Zustand, Redux, Tailwind CSS, ShadCN UI, Ant Design
Backend: Node.js, Express.js, Drizzle, Prisma ORM, Mongoose, Redis, BullMQ, Better Auth, Auth.js, Clerk
AI & ML: LangChain, LangGraph, Vercel AI SDK, OpenAI API, Claude API, Gemini API, LangSmith
Databases: PostgreSQL, MongoDB, MySQL, Supabase, Neon.tech
DevOps & Cloud: AWS (EC2, ECS), Docker, GitHub Actions, DigitalOcean, Vercel, CI/CD Pipelines
Tools: Git, GitHub, Postman, Linux

Professional Experience

Standard Insights	Oct '24 – Present
Full Stack AI Engineer	Remote New York, USA
– Architected and shipped a multi-phase, multi-LLM AI report generation pipeline powering market trend analysis, consumer persona synthesis, and strategic insights – built as 32 durable Inngest functions with automatic retries, concurrency control, and failure callbacks, replacing a brittle single-request model and eliminating job loss on server restarts. – Designed and owned a distributed background worker microservice processing AI workloads across 10+ concurrent queue types (quantitative data analysis, interview translation, persona generation, B2B scraping, Google Sheets sync); implemented batching and per-survey concurrency locks to prevent race conditions at scale. – Built an AI-powered Survey Builder Engine with 7-intent discriminated-union Zod schema validation against Gemini structured output, enabling natural-language survey authoring and real-time LLM-to-state mutation with zero schema-drift between the API and database layer. – Engineered a 6-step autonomous AI agent pipeline that ingests user-uploaded Excel/CSV files from AWS S3, validates format, parses multi-sheet structures, runs LLM-based schema analysis via Gemini, extracts survey questions and respondents, and re-uploads a normalized output file - all as idempotent, memoized steps with 3-retry fault tolerance. – Built and optimized RESTful APIs across 50+ Express modules using MongoDB (50+ Mongoose models) and Redis caching, improving session state management and data-layer response consistency; implemented JWT-based auth, role-based access control, and Google OAuth2 token refresh lifecycle for third-party integrations (Google Sheets, Slack). – Containerized 3 production services (client, server, admin) with Docker and managed deployments on AWS ECS, Railway, integrating CI/CD pipelines via GitHub Actions; deployed the worker service as an AWS Lambda function for cost-optimized serverless execution at scale. – Delivered consumer intelligence dashboards and survey creation UIs using Next.js 15 App Router, TypeScript, TanStack Query, Zustand, and ShadCN/Radix UI, contributing to a Turborepo monorepo spanning 4 apps and 19 shared packages, reducing cross-app code duplication by an estimated ~40%.	
HJ Animation Shop LTD	Jul '23 – Sep '24
Full-Stack Developer	Remote Loughton, United Kingdom

- Built autonomous AI customer support platform (Aximus) using Next.js and Vercel AI SDK – engineered a ToolLoopAgent with dynamic tool-calling and a keyword-scoring RAG engine (no embedding API; 98% accuracy), with HMAC-SHA256 Meta webhook ingestion and atomic DB idempotency locks.
- Architected multi-tenant SaaS platforms in Next.js 16 and Drizzle ORM, including a floating-point-free commission math engine, custom binary JPEG/PNG metadata parser for AI portrait editing, and randomized exam engine with practical video submission grading.
- Integrated Better Auth with RBAC, 6-digit OTP 2FA, and session revocation across products; wired SumUp and UddoktaPay v2 payment flows, and serverless jsPDF routes with font-scaling for dynamic certificate generation.
- Delivered headless CMS (Parocia) with React Server Components (zero client JS on public pages), dynamic sitemap/robots.txt/JSON-LD schema compilation, UUID-hashed file storage with path-traversal protection.

Projects

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| Visual Source Real-time soccer match operations and social graphic automation | Live 2026 |
| <ul style="list-style-type: none"> – Architected a multi-tenant soccer club dashboard using Next.js 16, Turborepo, and Better-Auth, securing isolated workspaces for multiple sports organizations. – Engineered a headless canvas rendering pipeline with Fabric.js, node-canvas, and Sharp to compile dynamic matchday graphics stored in Cloudflare R2. – Integrated Facebook, Instagram, and Twitter Graph APIs, executing a polling and media chunking pipeline for real-time automated matchday posting. – Designed a resilient Sportmonks API data synchronizer using PostgreSQL advisory locks to prevent database write collisions during concurrent syncing. – Developed a live event logging UI with game-state tracking, automatic score override mechanisms, and a cron-scheduled graphic publishing queue. – Deployed a containerized production stack using Docker Compose, orchestrating PostgreSQL, Next.js, and Caddy to automate SSL/HTTPS certificates and reverse proxying. | |
| Everything Green Digital Sustainability & SEO Platform | Live 2025 |
| <ul style="list-style-type: none"> – Designed a Retrieval-Augmented Generation (RAG) chatbot using MongoDB and OpenAI embeddings, featuring dynamic HTML scraping, semantic chunking, and context-aware query optimization to deliver digital sustainability advice. – Developed an AI SEO recommendation engine that matches Google Ads suggestions to scraped website pages via cosine similarity, classifying conversion performance using a custom-trained CatBoost model. – Implemented a secure subscription billing system using the Factory Pattern to integrate Stripe and PayPal, supporting customer portals, dynamic coupons, and webhook validation. – Implemented a dynamic website badge generation system using Puppeteer-based headless rendering, dynamically generating customized carbon certification badges from HTML templates and real-time site scores. – Built a responsive Next.js 15 dashboard with interactive D3/Sankey data visualizations and developed a Manifest V3 Chrome Extension to inject performance auditing metrics directly into Google SERPs. | |
| Career Dock A Modern Job tracking application | Live 2024 |
| <ul style="list-style-type: none"> – Built an end-to-end Next.js and Express.js web application utilizing Redux Toolkit (RTK Query) for cached state management, reducing client-side API latency by 30%. – Integrated the Gemini API to construct prompt-chaining workflows that generate tailored resumes, LinkedIn profiles, and cover letters based on target job descriptions. – Architected secure authentication and subscription monetization pipelines using Clerk for multi-tenant identity verification and Lemon Squeezy for premium plan billing. – Designed interactive analytical dashboards using real-time charting components powered by dynamic Mongoose aggregations to track job application success rates. – Deployed the distributed architecture to Vercel and cloud servers with a fully automated CI/CD pipeline, ensuring zero-downtime releases and instant preview deployments. | |

Education

University of the People	2021 – 2024
Bachelor of Science in Computer Science	DEAC California, USA
– Graduated with CGPA 3.67	
– Enlisted for Dean's Award List	

Certifications

IBM: Full Stack Software Developer Specialization
Meta: Front-End Developer Specialization | Back-End Developer Specialization
Google: IT Support Specialization
Amazon Web Services: AWS Fundamentals Specialization

Languages

English: (Comfortable - professional and technical communication)
Bengali: (Native)
Hindi: (Communicative)